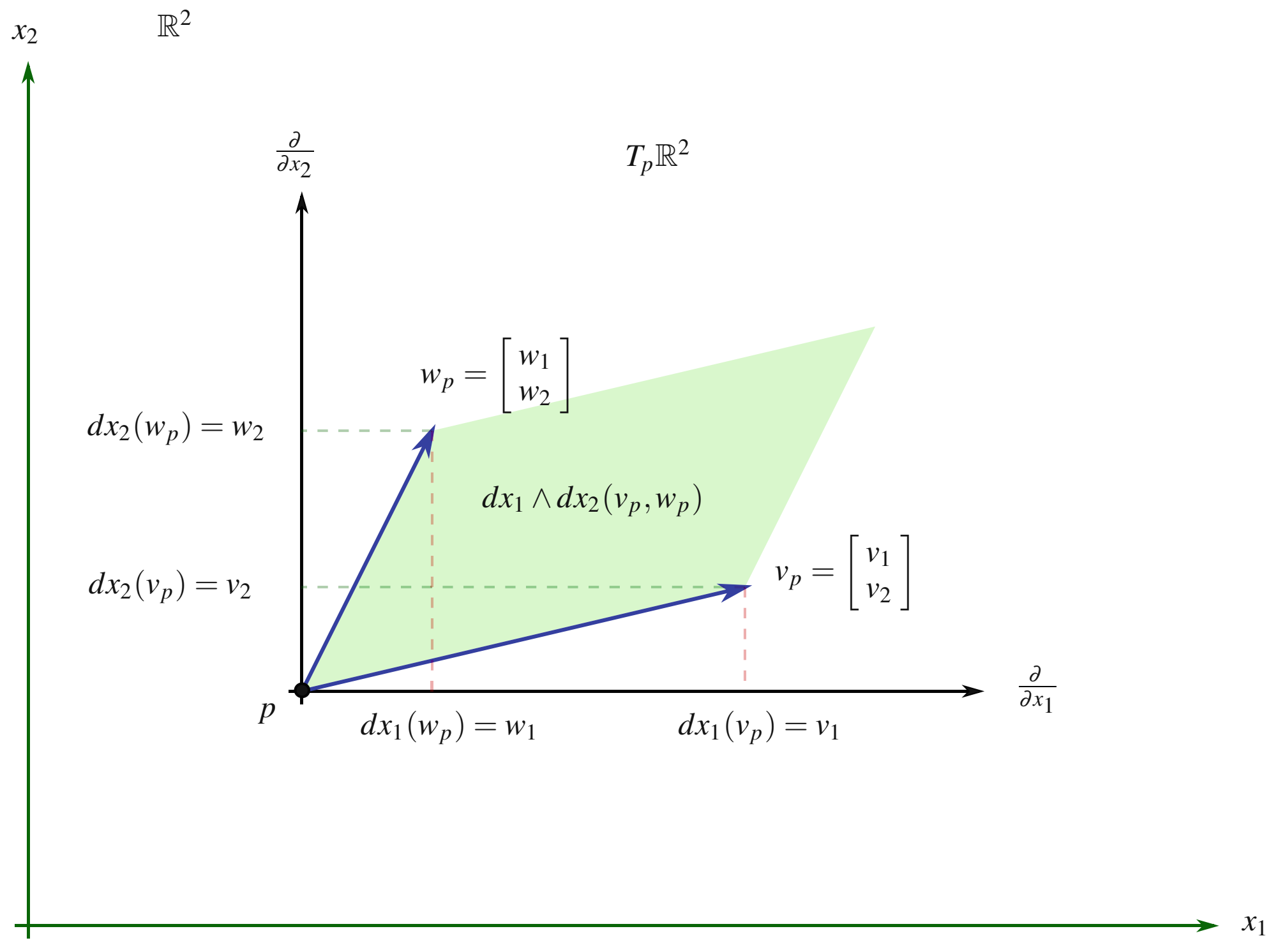

x_2
$\mathbb{R}^2$
$\frac{\partial}{\partial x_2}$
$T_p\mathbb{R}^2$
$w_p = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$
$dx_2(w_p) = w_2$
$dx_1 \wedge dx_2(v_p, w_p)$
$dx_2(v_p) = v_2$
$v_p = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$
$\frac{\partial}{\partial x_1}$
p
$dx_1(w_p) = w_1$
$dx_1(v_p) = v_1$
x_1